

Practice of Joins

1. Departments Table

```
CREATE TABLE Departments (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50) NOT NULL  
);
```

2. Students Table

```
CREATE TABLE Students (  
    student_id INT PRIMARY KEY,  
    name VARCHAR(50) NOT NULL,  
    dept_id INT,  
    FOREIGN KEY (dept_id) REFERENCES Departments(dept_id)  
);
```

3. Courses Table

```
CREATE TABLE Courses (  
    course_id INT PRIMARY KEY,  
    course_name VARCHAR(100) NOT NULL,  
    credits INT  
);
```

4. Enrollments Table

```
CREATE TABLE Enrollments (  
    enrollment_id INT PRIMARY KEY,  
    student_id INT,  
    course_id INT,  
    FOREIGN KEY (student_id) REFERENCES Students(student_id),  
    FOREIGN KEY (course_id) REFERENCES Courses(course_id)  
);
```

Data Insertion

```
INSERT INTO Departments VALUES (1, 'Computer Science');
```

```
INSERT INTO Departments VALUES (2, 'Information Technology');
```

```
INSERT INTO Departments VALUES (3, 'Software Engineering');
```

```
INSERT INTO Departments VALUES (4, 'Electrical Engineering');
```

```
INSERT INTO Departments VALUES (5, 'Mathematics');
```

```
INSERT INTO Students VALUES (101, 'Alice Johnson', 1);
```

```
INSERT INTO Students VALUES (102, 'Bob Smith', 2);
```

```
INSERT INTO Students VALUES (103, 'Charlie Brown', 1);
```

```
INSERT INTO Students VALUES (104, 'Diana Prince', 3);
```

```
INSERT INTO Students VALUES (105, 'Edward Norton', 4);
INSERT INTO Students VALUES (106, 'Fiona Green', 2);
INSERT INTO Students VALUES (107, 'George Harris', 5);
INSERT INTO Students VALUES (108, 'Hannah White', 3);
INSERT INTO Students VALUES (109, 'Ivan Black', 1);
INSERT INTO Students VALUES (110, 'Julia Roberts', NULL);
INSERT INTO Courses VALUES (201, 'Database Systems', 3);
INSERT INTO Courses VALUES (202, 'Data Structures', 3);
INSERT INTO Courses VALUES (203, 'Operating Systems', 3);
INSERT INTO Courses VALUES (204, 'Computer Networks', 3);
INSERT INTO Courses VALUES (205, 'Web Development', 2);
INSERT INTO Courses VALUES (206, 'Artificial Intelligence', 4);
INSERT INTO Courses VALUES (207, 'Software Engineering', 3);
INSERT INTO Courses VALUES (208, 'Cyber Security', 3);
INSERT INTO Enrollments VALUES (301, 101, 201);
INSERT INTO Enrollments VALUES (302, 101, 202);
INSERT INTO Enrollments VALUES (303, 101, 205);
INSERT INTO Enrollments VALUES (304, 102, 201);
INSERT INTO Enrollments VALUES (305, 102, 203);
INSERT INTO Enrollments VALUES (306, 103, 202);
INSERT INTO Enrollments VALUES (307, 103, 204);
INSERT INTO Enrollments VALUES (308, 103, 206);
INSERT INTO Enrollments VALUES (309, 104, 201);
INSERT INTO Enrollments VALUES (310, 104, 207);
INSERT INTO Enrollments VALUES (311, 105, 203);
INSERT INTO Enrollments VALUES (312, 105, 208);
INSERT INTO Enrollments VALUES (313, 106, 205);
INSERT INTO Enrollments VALUES (314, 106, 206);
```

INSERT INTO Enrollments VALUES (315, 106, 207);

INSERT INTO Enrollments VALUES (316, 107, 202);

INSERT INTO Enrollments VALUES (317, 108, 201);

INSERT INTO Enrollments VALUES (318, 109, 206);

INSERT INTO Enrollments VALUES (319, 109, 208);

Tasks:

- You have two tables, Students(student_id, name, dept_id) and Departments(dept_id, dept_name). Write a query to display each student's name along with their department name.
- Display all students and their enrolled course names. If a student is not enrolled in any course, still show their name with NULL for the course.
- Display the names of students who are enrolled in the course "Database Systems".
- Display student names, their department, and the courses they are enrolled in.
- Find students who are enrolled in more than 2 courses and display their name, department, and course count.
- Display the top 5 students with the highest number of course enrollments. Show their name and total courses.
- Display the top 3 courses (with at least 2 enrolled students) that have the highest enrollment count, but only consider courses worth 3 or more credits.